

Title: Implementation of K-Nearest Neighbors (KNN) for Classification

1. Dataset Source

The dataset used in this experiment is the **Breast Cancer Wisconsin Dataset**, which is commonly used for machine learning classification tasks.

Dataset Link (Kaggle):

<https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data>

This dataset is widely used to classify tumors as **malignant (cancerous)** or **benign (non-cancerous)** based on cell features.

2. Dataset Description

The **Breast Cancer Wisconsin Dataset** contains diagnostic measurements of breast cancer tumors.

Dataset Characteristics

- Total Instances: **569**
- Total Features: **30**
- Target Classes: **2**
- Type: **Binary Classification**

Important Features

Some important attributes in the dataset include:

Feature	Description
radius_mean	Mean of distances from center to points on the perimeter
texture_mean	Standard deviation of gray-scale values
perimeter_mean	Mean size of the tumor perimeter
area_mean	Mean area of tumor

smoothness_measures Variation in radius lengths
n

Target Variable

Target	Description
M	Malignant (Cancerous)
B	Benign (Non-cancerous)

The goal is to predict whether a tumor is **malignant or benign** based on these features.

3. Mathematical Formulation of the Algorithm

The **K-Nearest Neighbors** algorithm is a supervised learning technique used for classification and regression.

It works by finding the **K nearest data points** to a new input sample and assigning the most frequent class among those neighbors.

Euclidean Distance Formula

Distance between two points is calculated as

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

Where:

- (x_i) = feature value of the test sample
- (y_i) = feature value of the training sample
- (n) = number of features

Classification Rule

```
[  
Class(x) = Majority\ Vote(K\ Nearest\ Neighbors)  
]
```

If **K = 5**, the algorithm checks the **5 nearest neighbors** and predicts the most common class.

4. Algorithm Limitations

Limitations of **K-Nearest Neighbors** include:

- **High computational cost** for large datasets because distance must be calculated for all points.
 - Sensitive to **irrelevant or noisy features**.
 - Performance decreases with **high dimensional data** (curse of dimensionality).
 - Requires **feature scaling** because distance calculations depend on feature magnitude.
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5. Methodology / Workflow

The following steps were performed in the experiment:

Step 1: Data Collection

The **Breast Cancer Wisconsin Dataset** was obtained from Kaggle.

Step 2: Data Preprocessing

- Dataset loaded using Pandas
- Unnecessary columns removed
- Target labels encoded
- Feature scaling applied using StandardScaler

Step 3: Train-Test Split

Dataset split into:

- **80% Training Data**
- **20% Testing Data**

Step 4: Model Training

The **K-Nearest Neighbors** classifier was trained using the training dataset.

Step 5: Model Prediction

The trained model predicted tumor classes for the test dataset.

Step 6: Model Evaluation

Model performance was evaluated using:

- Accuracy Score
 - Confusion Matrix
 - Classification Report
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6. Performance Analysis

The performance of the model was analyzed using evaluation metrics.

Accuracy

Accuracy measures how many predictions were correct.

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Confusion Matrix

The confusion matrix shows:

- True Positive
- True Negative
- False Positive
- False Negative

Example result:

Model	Accuracy
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KNN ~0.95

The **K-Nearest Neighbors** model performed well in classifying tumors based on their diagnostic features.

7. Hyperparameter Tuning

Hyperparameter tuning helps improve the performance of machine learning models.

Important parameters for **K-Nearest Neighbors** include:

Parameter	Description
n_neighbors	Number of nearest neighbors (K value)
metric	Distance metric (Euclidean, Manhattan)
weights	Uniform or distance-based weights

Example:

```
KNeighborsClassifier(n_neighbors=5)
```

Different values of **K** were tested, and the value that gave the best accuracy was selected.

Conclusion

In this experiment, the **K-Nearest Neighbors** algorithm was implemented to classify tumors using the **Breast Cancer Wisconsin Dataset**.

The dataset was preprocessed and divided into training and testing sets. The KNN model predicted the class of tumors based on the similarity between data points using the Euclidean distance metric. Model performance was evaluated using accuracy, confusion matrix, and classification report.

The results showed that the KNN model achieved high accuracy in distinguishing between malignant and benign tumors. This experiment demonstrated that KNN is a simple yet effective classification algorithm for medical diagnosis and other pattern recognition tasks.